



1.07 Cement Procedure

1. Purpose

The purpose of the Standards Operating Procedure is to identify the Hazards and define the steps for doing Cement Jobs .

2. Scope

This procedure is intended for use by the Silverpoint employees .

3. Hazard Identification

- Physical Stress – Reduced Visibility
- Physical Stress – Ergonomics
- Mental Stress
- Equipment Failure – Stress
- Exposure to Corrosive Substance
- Potential Energy – Pressurized Liquids
- Slips, Trips and Falls
- Lifting, Twisting, Pulling and Pushing
- Pinch Points
- Burns
- Information Overload

4. EHS Requirements

- Fire Retardant Coveralls .
- Steel Toed Boots .
- Safety Glasses .
- H₂S Personal Monitor .



- Gloves
- Hard Hat

5. **References**

- Silverpoint Corporate Codes of Practice
- Alberta OH&S
- Silverpoint Task List
- Silverpoint THA 1.07

6. **Procedure**

6.1. **Back your truck into position .**

6.1.1. When backing into position , it is Silverpoint policy that when backing up you should have a Spotter (SOP 2.01), whether they are just making sure no one walks behind your truck or guiding you back . Most of the time you back in on the same side as the flock tanks (for surface and main hole). If they want you anywhere else they would let you know during the safety meeting .

6.2. **Attend the Pre Cement Job Safety Meeting .**

6.2.1. It is always good policy to attend the safety meetings . You can get a lot of information from the safety meetings . They can tell you how much cement you should be getting , when , how strong the cement is (calcium content) and how long the cement job should take . At these meetings ask questions about the job (i.e. Dye or no dye, De-foamer added).

6.2.2. If you are new to cement jobs , **let them know** , that way you are not as nervous and they can keep an eye on you, in case you need a hand with anything .

6.2.3. They will also tell you at these safety meetings what to do if you see something that isn't right, or something leaking , during the job .

6.3. **Connect your Hose to the Truck and run it to the Cellar or the Shaker Box, depending on the job.**

6.3.1. Refer to SOP 1.04 for instructions on this step .

6.3.2. It is also a good policy to tie your hoses down so if the hose kicks that you still have control of it.



6.3.3. Make sure that your truck is working properly ; you are on suction and have 20 inches of suction . **Have everything ready before the Cement Job Safety Meeting .**

6.4. **Check the Shaker Box or the Cellar .**

6.4.1. While the cementers are getting ready, you have some time to check the shaker box to make sure that it is cleaned out (to avoid plugging your hose).

6.4.2. If you are doing a surface cement job, you can check the cellar, to make sure that there aren't any big rocks or garbage that could potentially plug your hose .

6.4.3. Before any cement jobs, it is good policy to shale out your truck, this way you know that the truck is empty and the gauge won't give you a false reading while sucking on cement . The tank is easier to clean when shaled out. The cement might harden on the shale and this will cause problems later on.

6.5. **Loading Cement .**

6.5.1. On most shallow gas wells, there are two cement jobs, surface and main hole. On deeper wells there can be three, surface , intermediate or main hole .

6.5.2. Use a hose hook or rope tied to the vacuum hose to prevent back strain during surface cement in the cellar .

6.5.3. When doing cement jobs, the surface is done usually from the cellar and the main hole is done from the shaker box. The rule of thumb is that you start loading when the CONSULTANT tells you too. He is the man in charge, sometimes the rig hands will get nervous and try and tell you to load to soon . Never just throw your hose into the shaker box or the cellar, put it in slowly, this is just in case there is something in there that might plug your hose . Ask for a water line to thin out the cement during a surface cement job.

6.5.4. De-foamer is used by the cementers to control the cement from foaming up. This does not always happen , but if it does you should realize that this will take up more room in your truck when you suck it on. The cement can foam very easily, if this happens it can go through your piping and pump . Make sure that you let someone know if you are getting full, they can stop, while you go dump . It is better if you quit early, go dump, and then come back and finish, then filling your system with foam cement . It will harden just like regular cement . Cementers should be adding de-foamer but sometimes they forget .

6.5.5. Make sure that you clean your whole system if the cement foams . Diesel flush the system , clean the final filter and cyclone really good, drain everything and suck your valves out.

6.5.6. **Do not top up your load with Mud before dumping Cement , this will contaminate the Cement Pit.**

6.6. **PPE**



- 6.6.1. When you get cement water or mud under your coveralls , it acts as an abrasive . The constant turning of your wrists against your sleeves can cause very, very small cuts in your skin . Once the skin is damaged it is more likely to develop a burn when exposed to even a small amount of chemical .
- 6.6.2. You need to wear the proper PPE to protect yourself . Rubber gloves that cover the lower part of your arms when you are working in the mud tanks will help prevent your sleeves from getting wet. These are available at any safety supply store , keep your receipt and submit it for reimbursement as part of you safety allowance .
- 6.6.3. If you do get cement water or mud on your sleeves wash your sleeves off with clean water and change your coveralls as soon as you can .

6.7. Adding Sugar to the Cement .

- 6.7.1. Sugar slows the cement from setting up, it does not stop it, just slows it down . It is always good policy to add a cup of sugar to the cement on your truck . If you have a long distance to travel to the cement pit or something happens on the way, this will give you some extra time before the cement starts to set.
- 6.7.2. If sugar is not available , a can of pop can be used instead , the sugar content of pop is very high and acts the same as straight sugar . Pop can be added to the truck by pouring it into the valve after sucking the cement (while still on suction) . A couple of cans per load will slow the set up process down .

6.8. Clean up the Cementers .

- 6.8.1. After the cement job is over, hook up to the cement crew's high pressure line, so they can wash their mixing tub and their lines . They use water to clean out with so this will mix with the cement in the truck and thin it out. This usually does not take that long . Some of the time, the cementers are on the other side of the rig, so you might have to drive around to the other side of the rig . Some cement trucks have hookups (4 inch) right on the side of the tub . **Take extra care when driving around the rig, watch for people walking around and the Power Line hanging down .**
- 6.8.2. The cementers might ask you to clean up some dry cement that may have spilled during the cement job, because the vacuum trucks are a wet system you will have to wet the cement down before loading it or the dry powder will flow thru the vacuum system and damage the Hibon vacuum pump .

6.9. Disconnect Hose .

- 6.9.1. Refer to the SOP 1.02 for proper instructions on this step .

6.10. Dump in the Cement Pit or Ring .



- 6.10.1. Move into position engage Main PTO and Agitator PTO
- 6.10.2. Remove cap and hook up hose (SOP 1.02)
- 6.10.3. When dumping into the cement pit or ring, refer to the SOP 5.02, section 6.2. This procedure gives you a list of things to watch out for when dumping at a pit.
- 6.10.4. Free flow the cement off the truck and run the vibrator while doing so. Free flowing will prevent pressure from pushing the cement into any shale that may be on the bottom of the vacuum tank.
- 6.10.5. When dumping in a cement ring, you can not pressure off, and watch when you are backing up as not to hit the ring.
- 6.10.6. To clean out the agitator line, spin the impeller and suck out the line. This **MUST** be done after every cement job, Spinning the agitator will clean the impeller while the 4-way is on vacuum. **To properly clean the Agitator Valve, Agitator Line (inside the tank) and the Jet Line, build up to -20 psi and open the Agitator Valve. Repeat multiple times if needed.**
- 6.10.7. **Make sure that you suck out all your rear valves after dumping cement, especially your agitator valve. Drain the Final Filter and Cyclone to ensure that there is no Cement in the Vac System.**
- 6.10.8. Remove hoses and cap up all rear valves.
- 6.10.9. Disengage main PTO and Agitator PTO.
- 6.11. **Squeegee the vacuum tank clean.**
 - 6.11.1. If you think there is still cement in your tank after dumping open the rear door and have a look in designated area. (sawdust pile or mix off bin)
 - 6.11.2. You will have to complete a confined space entry permit (SOP 4.04) before entering the tank. **If you are not going back to the rig to start cleaning tanks, you must open your tank and clean it out.** If you don't, the cement will harden and may damage the agitator.

Remember : if you are in doubt about what to do, call your Supervisor .